

OMKAR KARHALE

9960058218 | omkarhale258210@gmail.com | Chhatrapati Sambhajnagar, MH | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Full Stack Developer with 1.6 years of experience developing and improving production software. Skilled in Java, Python, Go, Redis, and AWS, with hands-on experience in performance optimization and multithreading.

CORE TECHNICAL SKILLS

Languages: Java, Python, Golang (Go)

Java & Core Concepts: Core Java, OOP, Collections, Multithreading, Data Structures & Algorithms, Java 8+

Backend & APIs: Spring Boot, Hibernate (JPA), JDBC, RESTful APIs, Microservices, Node.js

Python & AI/ML: Python, FastAPI, RAG (Retrieval-Augmented Generation), Vector Databases, MCP (Model Context Protocol)

AI-Assisted Tooling: Cursor, GitHub Copilot

Databases: MySQL, PostgreSQL, MongoDB

Messaging & Caching: Apache Kafka, Redis

Cloud & DevOps: AWS, Docker, CI/CD Pipelines, Containerization

Testing & Quality: JUnit, Postman, SonarQube, Swagger

Tools & Methodology: Git, GitHub, Maven, Jira, Agile/Scrum

PROFESSIONAL EXPERIENCE

Full Stack Developer | Ampcus Tech Pvt Ltd

Feb 2025 – Present

Learning Management System (LMS)

- Optimized API performance by implementing a Redis caching layer for frequently accessed course data and session tokens, improving response times and reducing database read load.
- Worked with Apache Kafka to enable event-driven delivery of announcements and notifications on the real-time dashboard, improving notification delivery over the earlier polling-based approach.
- Gained hands-on exposure to AWS (EC2, S3) for application deployment, using Docker containerization and CI/CD pipelines to improve deployment reliability across dev, staging, and production.

CMEGP (Government Scheme Platform)

- Built a Python OCR document verification module (OpenCV, EasyOCR) with FastAPI backend APIs to auto-extract Aadhaar and PAN card details, eliminating manual data entry for every applicant submission.
- Designed an Aadhaar/PAN document-type validator that instantly flags invalid or mismatched uploads with a clear error message, improving data accuracy and cutting back-and-forth on rejected submissions.
- Reduced Aadhaar/PAN data extraction time from 50 seconds to 10 seconds (5x faster) by adding upfront document validation, edge-case handling, and an optimized OCR processing pipeline.

Incentive Tracker

- Engineered the end-to-end authentication and authorization module for Incentive Tracker — a role-based incentive management platform built with React.js, Java Spring Boot, and PostgreSQL, used to track, calculate, and approve employee/partner incentives — implementing secure login, registration, and role/permission-based access control (RBAC).
- Integrated Multi-Factor Authentication (MFA) using Microsoft Authentication (Microsoft Identity Platform / Azure AD), adding a secure MFA challenge step to the login flow.
- Automated session lifecycle management — logout with proper session invalidation and token/cookie clearing (JWT/OAuth2), plus a scheduled background job to purge expired tokens from the database, keeping the token store clean and reducing stale-session risk.

EDUCATION

B.Tech — Electronics & Telecommunication Engineering

2020 – 2024

Deogiri Institute of Engineering & Management Studies (DBATU) | University CGPA: 7.2

ACHIEVEMENTS

- Solved 300+ LeetCode Data Structures & Algorithms problems, reflecting consistent daily practice. Most Recent Badge: 100 Days Badge (2026).